

Review

Challenges to implementing urban sustainable stormwater management from a governance perspective: A literature review

Xiu-Juan Qiao*, Anders Kristoffersson, Thomas B. Randrup

Department of Landscape Architecture, Planning and Management, Swedish University of Agricultural Sciences, Slottsvägen 5, Alnarp, 230 53, Sweden

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ABSTRACT

Conventional piped drainage systems are currently being challenged by climate change and urbanization, leading to pluvial flooding, with negative consequences for people's properties and lives. Sustainable stormwater management has been suggested as an alternative, but implementation of its infrastructures is often limited to demonstration zones. Governance factors are among the main reasons for the slow pace. A literature review analyzing 44 peer-reviewed papers was conducted to examine the challenges from a governance perspective. The challenges identified were summarized based on actors, resources, rules of the game, and discourses. Challenges such as unclear leadership and responsibilities, lack of funding, lack of cost data on sustainable stormwater management solutions, lack of space and knowledge, lack of uniform guidelines, and lack of stakeholder participation were found to be the most prominent reasons. Suggested solutions to overcoming these challenges include e.g., researchers acting as knowledge brokers, the involvement of social capital among local stakeholders, re-using vacant land, establishing uniform guidelines, and increasing market incentives to engage stakeholders. The governance structures of stormwater management were categorized as hierarchical, closed and open co-governance arrangements, while open co-governance arrangement was suggested as a useful governance structure. The difference is that the latter structure engages stakeholders, which is of benefits to increased sustainable stormwater management implementation. More research on social-economical aspects is suggested. These findings can improve understanding among decision makers and other relevant actors of challenges related to implementing sustainable stormwater management as a primary solution to the current climate and urbanization challenges.

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* Corresponding author.

E-mail addresses: xiujuan.qiao@slu.se (X.-J. Qiao), Anders.Kristoffersson@slu.se (A. Kristoffersson), thomas.randrup@slu.se (T.B. Randrup).

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1. Introduction

Cities around the world are suffering from pluvial flooding because of increased urbanization and climate change (Cousins, 2017a; Petit-Boix et al., 2017). A general increase in impervious surfaces and frequency of cloudbursts in urban areas increase stormwater runoff and lead to high peak flow events (Dhakal and Chevalier, 2016). The main problem seems to be that old and small-diameter combined sewers cannot cope with huge amounts of stormwater in a short time, leading to frequent pluvial flooding events, especially during summer. This has been proven to have detrimental effects on people's properties and lives (Dhakal and Chevalier, 2016). Moreover, overflow of mixed wastewater and stormwater leads to pollution of receiving water bodies and impairs general water quality. Thus, conventional stormwater management solutions based on combined piped drainage systems have been criticized as being unsustainable (Gao et al., 2016), because it does not solve the contemporary environmental problems, leading into detrimental economic and social effects.

Efforts have been made to develop alternative solutions to manage stormwater in a more sustainable way, resulting in concepts and technologies such as low impact development (LID) in the US (Ahiablame et al., 2013; Gogate et al., 2017; Pyke et al., 2011), water sensitive urban design (WSUD) in Australia (Bach et al., 2015; Wella-Hewage et al., 2016), sustainable urban drainage systems (SUDS) in the UK (Ellis and Lundy, 2016; Fryd et al., 2012), and the 'sponge city' in China (Jiang et al., 2017; Wang et al., 2017). These new solutions have been successfully implemented in many places, e.g., Portland in the US (Dhakal and Chevalier, 2016), Melbourne in Australia (Ferguson et al., 2013a, 2013b), and pilot sponge cities (Wuhan, Xi'an New Area, Zhenjiang, etc.) in China (Jiang et al., 2017). The core concept of these new solutions is using green infrastructure (Law et al., 2017), e.g., rain gardens, green roofs, green walls, wetlands, water ponds, etc., to infiltrate, detain, retain stormwater, and by mimicking nature to reduce influence of hydrological balance (Semadeni-Davies et al., 2008). Such approaches have recently been defined as Nature-based Solutions (NbS) (Laforteza et al., 2017) which can not only mitigate environmental pressure by reducing pluvial flooding, but also provide many other ancillary socio-economic and environmental benefits, such as providing more green space, improving air and water quality, improving biodiversity, decreasing urban heat island effects (environmental benefits), providing cost efficient solutions and increasing green jobs (economic benefits), and generally creating a coherent networked landscape to promote human resilience, mental health, recreational, wellbeing and social cohesion (social benefits) (MacKinnon et al., 2008; Eggermont et al., 2015; Meerow and Newell, 2017; Pauleit et al., 2017). In this paper, we refer to these new solutions as sustainable stormwater management (SSM): using green infrastructure as NbS to mitigate and manage stormwater, and simultaneously provide multiple environmental, economic and social benefits. Originally, sustainable development was defined as "development that meets the needs of the present

without compromising the ability of future generations to meet their own needs" (UN, 1987). Here, in the concept of SSM, sustainability is described from a three-part focus: environmental, economic and social considerations, with environmental sustainability as the main focus. A vast amount of research has been conducted on the technical aspects of SSM in terms of: infiltration function, e.g., rain gardens (Chaffin et al., 2016; Church, 2015), green roofs (Mees et al., 2013), and wetlands (Schulte-Hostedde et al., 2007); reducing pollutants, e.g., phosphorus, heavy metals (Kaplowitz and Lupi, 2012; Rahman et al., 2015); and hydraulic models for hydraulic routing on-site (Dietz, 2007). In parallel with the many technical papers, an increasing number of research papers focusing on the governance aspects, e.g., policies, strategies, and decision-making frameworks, have been published during the past 10 years (Gogate et al., 2017; Goonrey et al., 2009; Martin et al., 2007).

However, to date, SSM practices seem to be limited to demonstration areas or pilot projects, despite a global focus and broad and generally accepted knowledge on how to install relevant solutions (de Graaf et al., 2009). The literature suggests that the slow pace of implementing SSM is mostly influenced by governance factors rather than technical aspects (Brown and Farrelly, 2009; Mankad et al., 2015). For example, clear organizational roles is essential for technological innovation (Lane et al., 2017), and support from regulatory authorities can increase the installation of rain tanks by private stakeholders (van der Sterren et al., 2009). Although governance factors are often identified as being critical, the various roles of the actors and the challenges they could encounter in implementing SSM are complicated and have not been studied systematically. Therefore, in this study we conducted a review of contemporary literature in order to identify the most important challenges to implementing an actual and factual change to SSM. Our aim was to provide a foundation for future research relating to SSM from a governance perspective and as reference for decision makers and other relevant actors.

2. Methods

2.1. Theoretical foundation

The policy arrangement model (PAM) is a conceptual framework, visualized as a tetrahedron (Fig. 1), developed in environmental policy studies to assist understanding stability of content and organization of a policy domain (Arts et al., 2006). In summary, a policy arrangement is defined as the state in which the interaction between political actors and resources and rules of the game solidifies in a temporary stable structure (institutionalization), before the driving force of evolution forces them to readjust their interdependency (political modernization). The PAM comprises four profoundly interconnected dimensions: actors, resources, rules of the game, and discourses. Each of these dimensions affects the others and changes the shape of the entity, such as new actors' appearance may lead to division of resources, new rules of the

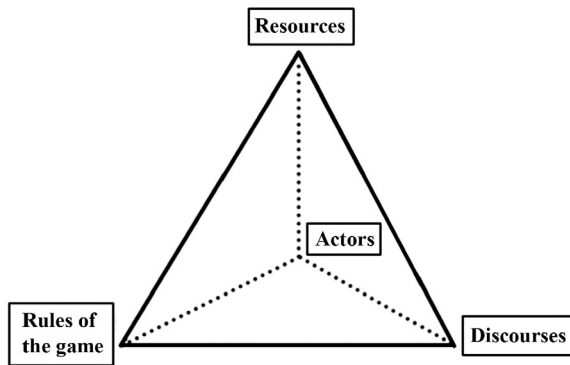


Fig. 1. The policy arrangement model (PAM) (adapted from Arts et al., 2006).

game and/or new discourses.

In the specific context of SSM, actors refer to individuals or organizations involved e.g., landscape architects, urban planners, water professionals from the municipalities, stakeholders (like schools, private landowners) etc., while resources include many types, e.g., financial resources, knowledge, skills, products, facilities and land priorities. Rules of the game are the current formal and informal norms for actors when they determine actions on implementation of SSM. Discourses refer to attitudes and perspectives of implementation of SSM. Generally, local government is responsible for managing stormwater drainage systems with grey infrastructure from planning to maintenance (Morison and Brown, 2010). The governance structure is hierarchical, with national government setting strategies and local government implementing centralized stormwater management systems, particularly those managed by municipal water departments. Thus, the conventional piped drainage systems can be seen as an initial policy arrangement that is being challenged by the emergence of SSM. From an SSM perspective, it requires devolved governance structures with decentralized management by cooperation across a variety of departments, e.g., water, planning, parks and environmental departments, as well as involving stakeholders at local or neighborhood levels and increasing efficient interactions between government actors and stakeholders. The policy arrangement model can help to understand the influence factors of SSM implementation.

Governance, as a concept, is popularly used in social and political science, but there is no uniform definition. It generally refers to a policy process of the organizational ways or the “interaction between different social and political actors” (Kooiman, 2003), which

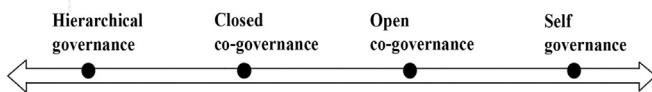


Fig. 2. The four governance modes (adapted from Arnouts et al., 2012).

aims to promote the policies for public purposes. Kooiman (2003) suggests three governance modes (hierarchical governance, co-governance, and self-governance) to illustrate modern governance structures. However, it is difficult to differ and define interactions between governmental and non-governmental actors in the co-governance mode. Arnouts et al. (2012) expanded the three modes into four, namely hierarchical governance, closed co-governance, open co-governance, and self-governance (Fig. 2). The governance modes shift in terms of the active actors in a specific period because of the changeable nature of resources. To allow analysis of governance shifts, the four governance modes can be connected with the four aspects of the PAM to address the changeability of governance (Table 1). In a hierarchical governance arrangement, the primary actors are mainly from the local government and holding power to decide the use of resources and force the actions. In a closed co-governance arrangement, strictly selected actors including governmental and non-governmental actors cooperate based on specific goals and powers pooled. In an open co-governance arrangement, many different types of actors conduct actions by devising power and cooperate flexibly based on their resources. In a self-governance arrangement, non-governmental actors play a leading role in distributing resources and decision making.

2.2. Systematic search

Initially, two autonomous researchers conducted a systematic literature search using the same search string: Urban AND stormwater* NEAR/5 manage* in Web of Science, and Urban AND (stormwater* W/5 manage*) in Scopus. “AND” in the search block is used to combine search words and thus to narrow down the results. Asterisks are used to make sure all the papers including the words “stormwater” and “manage” will be searched. “NEAR/5” in Web of Science or “W/5” in Scopus is to narrow down the search results by searching papers with “stormwater*” and “manage*” appearing within five words. The basic criteria for the literature search were year, language and peer-reviewed. Thus, only peer-reviewed papers written in English were sought within the period from 2007 to May 2017, since initial searches indicated that the amount of papers focusing on governance perspectives has increased greatly during the past 10 years. As a second step, the articles were limited by reading titles and abstracts, based on the four policy arrangement terms: ‘actors’, ‘resources’, ‘rules of the game’, and ‘discourses’:

- The actors (e.g., landscape architects, urban planners, water professionals, stakeholders)
- The resources (e.g., financial, land use, knowledge)
- The rules of the game (governance structures, policy arrangements, decision-making, implementation strategies)
- The discourses (attitudes, awareness, understanding, perspectives of actors).

Each of the researchers combined the search results from Web of Science and Scopus and deleted duplicates, resulting in 133 (researcher 1) and 170 (researcher 2) relevant papers (104 papers

Table 1

Four governance modes combined with PAM (adapted and further developed from Arnouts et al., 2012).

PAM	Governance Arrangements	Closed Co-governance	Open co-governance	Self-governance
Actors	Mainly government actors	Selected mixed group of actors	Large mixed group of actors	Mainly non-government actors
Resources	Power with government	Power pooled	Power diffused	Power with non-government
Rules of the game	Government coercion	Restricted cooperation	Flexible cooperation	Non-government forerunning
Discourses	Reluctant to change to initiatives	An initiative as an auxiliary solution	An initiative as an alternative solution	An initiative as a dominant solution

were duplicates, while 95 papers were different). The two researchers then read the abstracts of all papers again and together identified 125 potentially relevant papers. Full texts were retrieved from the authors for papers without full texts in the search engines. In total, 101 papers with full text and 24 without were retrieved. As the third step, the 101 papers were scrutinized. Those with a primary focus on water systems, stormwater quality, pollution decrease, stormwater runoff, peak-flow modeling, and SSM

solutions presented primarily as technical practices were excluded. A total of 42 papers then remained. During the analysis process, newly published papers based on the search criteria were sought and two papers were added to the list reviewed. Thus, in total, 44 papers were reviewed. The search process is shown in a diagram in Fig. 3, and the arrows in the figure show the narrow down process of the search results.

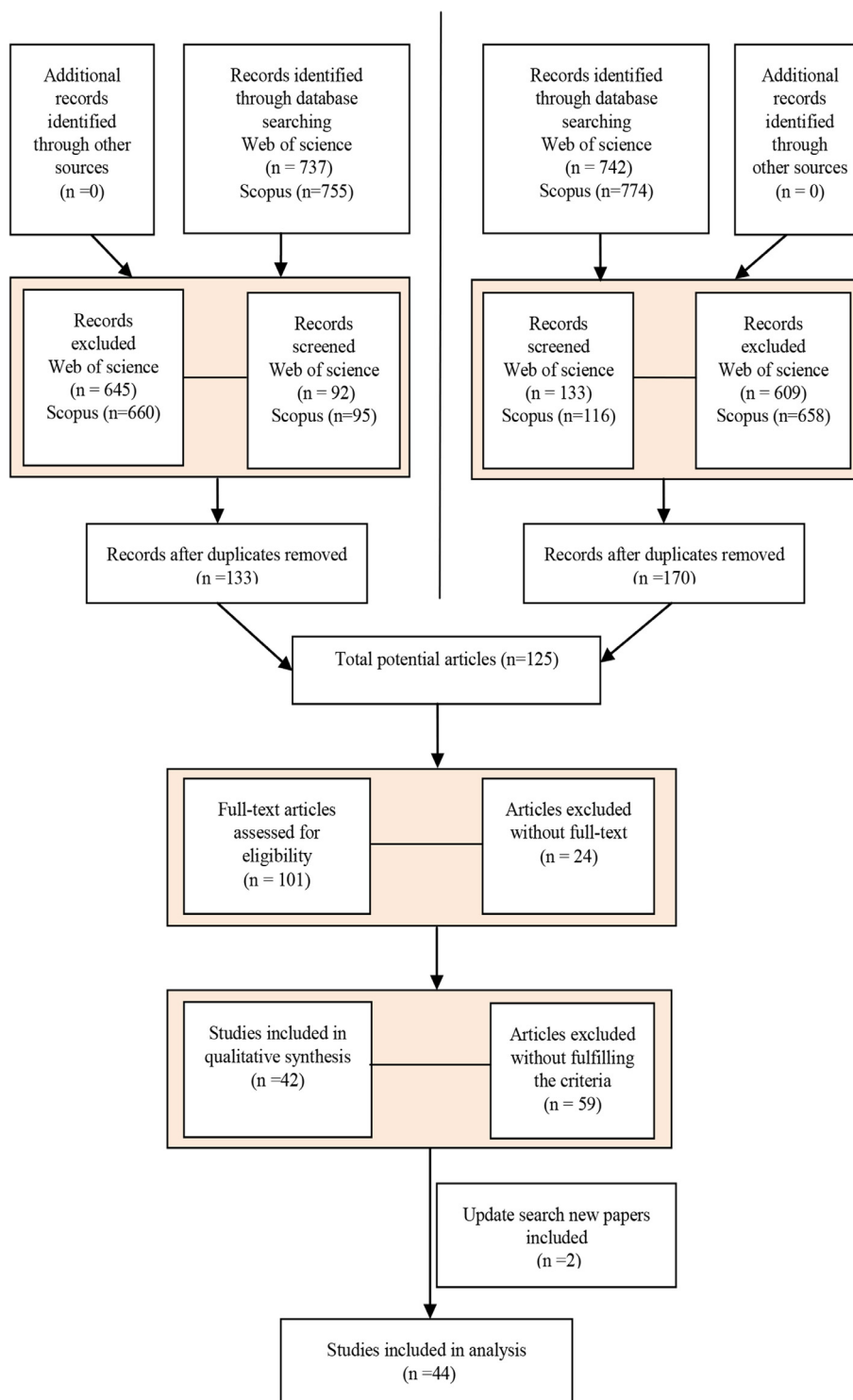


Fig. 3. Search process diagram for relevant publications.

3. Findings and discussion

The challenges to implementing SSM were summarized in terms of the four aspects mentioned above: actors, resources, rules of the game, and discourses (Table 2).

In the following sections of 3.1–3.3, we discuss the findings listed in Table 2. As mentioned in section 2.1, PAM (Fig. 1) is not a static model and the four aspects are interrelated. The aspect of discourses has been defined as “the views and narratives of the actors involved in terms of norms and values, definitions of problems and approaches to solutions” (Arts et al., 2006). Therefore, the challenges related to discourses are included under the headings of the other three aspects. After presenting the various challenges, we summarize these challenges in relation to the identified governance arrangements and the different sustainability considerations.

3.1. Actors

3.1.1. Unclear leadership and responsibilities

Unclear leadership is frequently mentioned as a challenge to SSM implementation (Burns et al., 2015; Cettner et al., 2014a; Cettner et al., 2013; Dhakal and Chevalier, 2016; Dhakal and Chevalier, 2017; Dolowitz et al., 2012; Keeley et al., 2013; Martin et al., 2007; Morison and Brown, 2010; O’Sullivan et al., 2012; Perales-Momparler et al., 2015; Porse, 2013; Prosser et al., 2015; Roy et al., 2008). It is beneficial to have champions or case front-runners that play a key role in adapting SSM, empowering the relevant actors to practice, and persuading policy makers to change or make new strategies and communicate with stakeholders (Brown et al., 2013). A study examining the process of implementation of green roofs in five case cities in the US stressed that local authorities should play a leading role and take the main responsibility for ensuring that adaptation actions are taken in early

stages of policy processes (Mees et al., 2013). Similarly, a study in Emscher and Berlin, Germany, highlighted the need for local authorities to assume leadership while enabling the participation of stakeholders in the implementing process (Nickel et al., 2014).

The local authority thus plays the leading role as an entity, but relevant actors within the local authority are also confronted with the problem of unclear leadership. Backhaus et al. (2012) explored the challenges facing landscape architects during their design process, and argued that landscape architects should play a leading role in the planning process by integrating water experts and urban planners. Others claim that urban planners should play a leading role and dedicate their efforts to the long-term planning process, rather than relying on consultancy companies (Desimini, 2013), and integrated planning can be used as a platform for the various actors to communicate and present their attitudes and perspectives (Nickel et al., 2014). Looking at the water professionals, by joining the planning process from the beginning or at a later stage, they can influence and encourage implementations of sustainable solutions (Cettner et al., 2013), but their early involvement cannot guarantee a change to sustainable alternatives. Barbosa et al. (2012) also suggested that the water department should dominate the planning process of SSM by cooperating with other departments. New policy and strategies or legislation may provide water professionals with opportunities to take a leading role in the planning process.

It is still unclear who should play the leading role in implementation of SSM, which can lead to unclear responsibilities (Burns et al., 2015; Dhakal and Chevalier, 2017; Dolowitz et al., 2012; Keeley et al., 2013; Martin et al., 2007; Morison and Brown, 2010; O’Sullivan et al., 2012; Perales-Momparler et al., 2015; Porse, 2013). For example, the responsibility for planning SSM is unclear within planning departments and water departments. In addition, involvement by private and other stakeholders leads to a fragmented stakeholder environmental network (e.g., land properties)

Table 2
Challenges to implementing SSM.

Aspects	Challenges
Actors	• Unclear leadership and responsibilities (Burns et al., 2015; Cettner et al., 2014a; Cettner et al., 2013; Dhakal and Chevalier, 2016; Dhakal and Chevalier, 2017; Dolowitz et al., 2012; Keeley et al., 2013; Martin et al., 2007; Morison and Brown, 2010; O’Sullivan et al., 2012; Perales-Momparler et al., 2015; Porse, 2013; Prosser et al., 2015; Roy et al., 2008) • Difficult to engage stakeholders (Dhakal and Chevalier, 2016; Perales-Momparler et al., 2015; Porse, 2013) and Difficult to cooperate with different stakeholders (Backhaus et al., 2012; Barbosa et al., 2012; Perales-Momparler et al., 2015) and Lack of public understanding of their role in stormwater management (Chaffin et al., 2016; Keeley et al., 2013) • Lack of institutional capacity (Perales-Momparler et al., 2015; Roy et al., 2008) • Lack of experienced expertise (Keeley et al., 2013; Morison and Brown, 2010) • Disagreement with the effectiveness or the means of achieving planning goals (Cousins, 2017b)
Resources	• Lack of funding (Barbosa et al., 2012; Burns et al., 2015; Cettner et al., 2014b; a; Chaffin et al., 2016; Dhakal and Chevalier, 2016; Dolowitz et al., 2012; Keeley et al., 2013; Morison and Brown, 2010; Perales-Momparler et al., 2015; Porse, 2013; Prosser et al., 2015; Roy et al., 2008) • Lack of evidence on what SSM costs and SSM efficiency (Backhaus et al., 2012; Burns et al., 2015; Cettner et al., 2014b; Chaffin et al., 2016; Cousins, 2017b; Dhakal and Chevalier, 2017; Flynn and Davidson, 2016; O’Sullivan et al., 2012; Perales-Momparler et al., 2015; Porse, 2013; Roy et al., 2008; Zhang et al., 2017; Prosser et al., 2015; van der Sterren et al., 2009) • Lack of effective market incentives (Cousins, 2017b; Prosser et al., 2015; Roy et al., 2008) • Lack of space (Barbosa et al., 2012; Cousins, 2017b; Keeley et al., 2013; O’Sullivan et al., 2012; Prosser et al., 2015; Zhang et al., 2017) • Lack of knowledge (Barbosa et al., 2012; Brown et al., 2013; Burns et al., 2015; Cettner et al., 2014a; Chaffin et al., 2016; Cousins, 2017b; Perales-Momparler et al., 2015; Porse, 2013; Prosser et al., 2015) • Lack of staff and time (Burns et al., 2015; Dhakal and Chevalier, 2017; Morison and Brown, 2010)
Rules of the game	• Lack of incorporation of SSM in legislative mandates (Cettner et al., 2014b; a; Cettner et al., 2013; Chaffin et al., 2016; Cousins, 2017b; Dhakal and Chevalier, 2016; Dhakal and Chevalier, 2017; Keeley et al., 2013; Morison and Brown, 2010; Nanekely et al., 2016; Perales-Momparler et al., 2015; Porse, 2013; Prosser et al., 2015; Roy et al., 2008) • Lack of SSM standards (Brown et al., 2013; Cettner et al., 2014b; Chaffin et al., 2016; Dhakal and Chevalier, 2017; Keeley et al., 2013; O’Sullivan et al., 2012; Roy et al., 2008; van der Sterren et al., 2009)
Discourses	• Fragmented environmental stakeholder network (Morison and Brown, 2010) • Perceived risk in cost and performance (Barbosa et al., 2012; Dhakal and Chevalier, 2017) • Lack of identifiable environmental values (Morison and Brown, 2010) • Lack of awareness of the adaptability of SSM technologies and policies (Dhakal and Chevalier, 2017; Dolowitz et al., 2012) • Engineering culture (Barbosa et al., 2012; Burns et al., 2015; Cettner et al., 2014a; Chaffin et al., 2016; Dhakal and Chevalier, 2016, 2017) and Resistance to change (Barbosa et al., 2012; Dhakal and Chevalier, 2017; Porse, 2013; Prosser et al., 2015; Roy et al., 2008) • Lack of congruence between political and hydrological considerations (Dhakal and Chevalier, 2016; Flynn and Davidson, 2016; Keeley et al., 2013)

(Morison and Brown, 2010), which contributes to fragmented responsibilities in terms of construction and maintenance (Cettner et al., 2014a; Dhakal and Chevalier, 2016; Perales-Momparler et al., 2015; Prosser et al., 2015; Roy et al., 2008). It is worth noting that being the leader means owing power; the leader is confronted by questions of how to allocate resources (e.g., finance, staff time), how to build long-term trust relationships among various actors, and how to achieve cooperation and collaboration with relevant actors.

3.1.2. Researchers as brokers

Researchers from universities or research institutions can act as brokers within the local authority including landscape architects, urban planners, water professionals, and between the local authority and private stakeholders. It has been found that civil experimentation by holding workshops or building pilot projects can improve the willingness and capacity of municipalities to implement SSM (Burns et al., 2015). Researchers with a leading role in experiments could help to build long-term relationships and partnerships among the relevant actors, thereby increasing their trust and confidence in each other and their technical knowledge of SSM. Such partnerships are crucial for a change to new technologies and long-term cooperation between researchers and practitioners can lead to sharing of institutional responsibilities and co-investment (Prosser et al., 2015). As a bonus, researchers could collect research data from workshops and projects in terms of designing and construction.

3.1.3. Stakeholder participation: challenges and solutions

Since urban land is often private (Dhakal and Chevalier, 2016), and is often related with various interests of different stakeholders, private stakeholder involvement in implementation of SSM has the potential to be significant. It can help to manage stormwater on-site and reduce stormwater runoff, and also to solve the challenges of land and funding shortages. To date, stakeholder participation is limited to demonstration projects and lack of trust seems to be the main problem, expressed as mutual mistrust between the local authority and stakeholders. The local authority often does not trust private stakeholders to manage stormwater because of their general lack of knowledge of SSM. Because of lack of knowledge and mistrust of local authority, stakeholders are often unwilling to take part in the practice of SSM (Bos and Brown, 2015).

Participation by private stakeholders needs economic incentives, e.g., costs-shared between private stakeholders and the local community or contract developers can increase the willingness to pay for installing rain barrels or building rain gardens (Green et al., 2012; Newburn and Alberini, 2016). In some cases, collecting an impact fee is also an effective way to increase the practice of SSM (Lu et al., 2013). Developers would have to pay an impact fee if they did not use SSM solutions. Education is another crucial aspect to encourage public participation. One good example is the USA national LID Atlas, which is a platform on Google Maps with detailed information on all LID practice projects (Dickson et al., 2011). The atlas is intended for use by municipal officials, contract developers, and private stakeholders to refer to the information when implementing relevant SSM solutions on their own land. It has also been found that combining an education campaign with cost incentives achieves better outcomes, and contributes to substantially reduced runoff, if a facility installation fee is paid by the local authority (Thurston et al., 2010).

Money is not the only factor influencing stakeholder participation (Brown et al., 2016). Working to understand the perspectives, preferences, inputs, and values of stakeholders concerning SSM can be a way to start engaging stakeholders and get their support (Kaplowitz and Lupi, 2012). A choice experiment survey conducted

in Sycamore Creek, Ingham County, Michigan, US, revealed that high level streambank naturalization was the option most preferred by local stakeholders (Kaplowitz and Lupi, 2012). It is also important to involve the public jointly in the decision-making process, e.g., an auction “engaged residents to voluntarily bid on installation of stormwater management practices on their property” (Thurston et al., 2010). Investigating willingness to pay for popular SSM solutions like rain gardens is another effective way of engaging the public in decision making (Chui and Ngai, 2016; Newburn and Alberini, 2016). Many other factors are also significant for increasing private stakeholders participation in SSM implementation, such as time needed to understand the concept of SSM, learn the technologies of SSM solutions, change attitudes to SSM (Bos and Brown, 2015), write clear information about SSM solutions on signs in accessible green spaces, and document the contributions and commitment of stakeholders in implementation of SSM (Gao et al., 2016).

3.2. Resources

3.2.1. Lack of funding

The funding problem is a crucial influencing factor, and has been ranked highest in list of impediments to implementation of SSM on watershed scale (Roy et al., 2008). Many researchers also argued that financial challenges are the greatest barrier to SSM implementation (e.g., (Barbosa et al., 2012; Burns et al., 2015; Cettner et al., 2014b; a; Chaffin et al., 2016; Dhakal and Chevalier, 2016; Dolowitz et al., 2012; Keeley et al., 2013; Morison and Brown, 2010; Perales-Momparler et al., 2015; Porse, 2013; Prosser et al., 2015). On the one hand, there is a lack of funds at national, regional and/or local government level or difficulties in the local government obtaining funding from the state or central government (Keeley et al., 2013). On the other hand, there are limitations on the use of public or social funding in SSM approaches according to existing legislation and regulations.

3.2.2. Lack of evidence on what SSM costs, SSM efficiency and lack of effective market incentives

Lack of detailed cost data and information about running SSM solutions including maintenance costs (Backhaus et al., 2012; Burns et al., 2015; Cettner et al., 2014b; Chaffin et al., 2016; Cousins, 2017b; Dhakal and Chevalier, 2017; Flynn and Davidson, 2016; O'Sullivan et al., 2012; Perales-Momparler et al., 2015; Porse, 2013; Roy et al., 2008; Zhang et al., 2017) has led many local authorities, contract developers, and private stakeholders to show unwillingness to spend money on integrating SSM solutions because of the perceived risk in cost and performance (Barbosa et al., 2012; Dhakal and Chevalier, 2017). The lack of effective market incentives further exacerbates the situation (Cousins, 2017b; Prosser et al., 2015; Roy et al., 2008). Limiting the use of governmental funding in promoting stakeholder participation also decreases uptake of SSM, which in turn increases the local government's financial burden if few private investors are involved. Integrating social capital or private sector investments is argued as being significant for SSM implementation (Green et al., 2012). Stormwater management fees or allowances, municipal green infrastructure bonds, or other incentive mechanisms can be ways to encourage private investors (Dhakal and Chevalier, 2017).

Lack of performance evaluations of SSM solutions is noticed by many researchers, e.g., (Chaffin et al., 2016; Cousins, 2017b; Perales-Momparler et al., 2015; Porse, 2013; Prosser et al., 2015; Roy et al., 2008; van der Sterren et al., 2009). For conventional piped drainage systems, it is easy to calculate the investment and evaluate the performance in terms of stormwater runoff. However, evaluation of SSM performance is more difficult and the benefits of SSM are less

well-known, e.g., SSM solutions based on green infrastructure can provide multiple benefits and multiple ecosystem services, in addition to managing stormwater (Law et al., 2017). When in some cases, these benefits are not considered, SSM solutions appear more expensive than conventional solutions, leading to resistance to change (Barbosa et al., 2012; Dhakal and Chevalier, 2017; Porse, 2013; Prosser et al., 2015; Roy et al., 2008). Formulation of new evaluation criteria is necessary for actors to be able to communicate and measure green infrastructure performance (Chaffin et al., 2016), which is necessary in order to change actors' attitudes to SSM.

3.2.3. Lack of space

The increase of SSM implementation is always challenged by lack of space (Barbosa et al., 2012; Cousins, 2017b; Keeley et al., 2013; O'Sullivan et al., 2012; Prosser et al., 2015; Zhang et al., 2017), because it needs much urban space. City development is increasingly dominated by urban densification, and in many countries public land in cities only occupies a very small proportion compared with private land, e.g., 65–75% of land in US cities is private residential land (Dhakal and Chevalier, 2016). Moreover, it is argued that many significant issues inherently need to be considered for public land rather than stormwater management, e.g., land for schools, hospitals, and recreation. Some researchers report a gap between reconstruction on vacant land and integration of SSM infrastructures (Desimini, 2013). Thus, use of vacant land for implementation of SSM is a potential co-beneficial way to deal with the land shortage problem. Others also pointed out the value of vacant land for SSM (Burns et al., 2015; Prosser et al., 2015). Additionally, taking Melbourne as a case, it has shown that the land shortage challenge can lead to innovative solutions for SSM, bringing new business opportunities (Prosser et al., 2015).

3.2.4. Lack of knowledge

Challenges related to lack of knowledge and experiences regarding SSM implementation are reported frequently (e.g., (Barbosa et al., 2012; Brown et al., 2013; Burns et al., 2015; Cettner et al., 2014a; Chaffin et al., 2016; Cousins, 2017b; Perales-Momparler et al., 2015; Porse, 2013; Prosser et al., 2015). This connects with challenges among actors, e.g., lack of institutional capacity for SSM (Perales-Momparler et al., 2015; Roy et al., 2008) and lack of expertise (Keeley et al., 2013; Morison and Brown, 2010), and discourses challenges, e.g., lack of ability to identify environmental values (Morison and Brown, 2010) and lack of awareness of SSM solutions and the adaptability of SSM technologies and policies (Dhakal and Chevalier, 2017; Dolowitz et al., 2012). Some also mentioned a lack of analysis of available knowledge (Dolowitz et al., 2012). Transfer of knowledge and exchange of experience is usually based on workshops or conferences or via websites in relation to successful projects. When the collected experiences and knowledge are applied to other specific places, problems arise from weak couplings between knowledge and local conditions. Therefore, lack of localized SSM solutions to suit local contexts is mentioned (Barbosa et al., 2012; Dolowitz et al., 2012; Flynn and Davidson, 2016; Martin et al., 2007). The failure of practice leads to lack of confidence in SSM solutions. As a result, mistrustful attitudes lead to reluctance to change (Barbosa et al., 2012; Dhakal and Chevalier, 2017; Porse, 2013; Prosser et al., 2015; Roy et al., 2008), e.g., to new policy or strategies and new legislation (Church, 2015). Key staff turnover, lack of staff time (not enough resources input), and a shortage of design and installation staff or staff with interdisciplinary skills may also need to be considered for slow pace of implementing SSM (Burns et al., 2015; Dhakal and Chevalier, 2017; Morison and Brown, 2010).

3.3. Rules of the game

3.3.1. Lack of incorporation of SSM in legislative mandates

Existing regulations, plans, and legislation tend to not include implementation of SSM solutions (Dhakal and Chevalier, 2017; Keeley et al., 2013; Porse, 2013) and legal authorities have to follow the existing stormwater management approach. Thus, lack of organizational support (Cettner et al., 2014b; a; Cettner et al., 2013; Chaffin et al., 2016; Perales-Momparler et al., 2015; Prosser et al., 2015) and lack of legislative mandates (Cettner et al., 2013; Chaffin et al., 2016; Cousins, 2017b; Dhakal and Chevalier, 2016; Nanekely et al., 2016; Roy et al., 2008) constrain the increase of implementing SSM. Removing the requirements for implementing piped drainage systems on the existing legislative mandates has been recommended (Dhakal and Chevalier, 2017).

Since there are no accessible structures for the actors to build inter-legitimacy in local authorities (Morison and Brown, 2010) and no regulatory frameworks (Perales-Momparler et al., 2015), it is difficult to change the engineering culture and mindset (Barbosa et al., 2012; Burns et al., 2015; Cettner et al., 2014a; Chaffin et al., 2016; Dhakal and Chevalier, 2016, 2017). Restructuring of the general governance structure of local government would be a useful way forward, as discussed in section 3.4.

3.3.2. Lack of SSM standards

Lack of standards and guidelines for planning, constructing, and maintaining SSM is viewed as an important challenge (e.g., (Brown et al., 2013; Chaffin et al., 2016; Keeley et al., 2013; O'Sullivan et al., 2012; Roy et al., 2008). This shortcoming is partly because actors disagree with the effectiveness of SSM and ways of achieving planning goals (Cousins, 2017b), and partly because of the unaddressed problem of lack of congruence between hydrological and political boundaries (Dhakal and Chevalier, 2016; Flynn and Davidson, 2016; Keeley et al., 2013). Establishing uniform standards and guidelines for design, construction, and maintenance is critical for acceptance and implementation of SSM, but the local context should not be neglected when selecting specific SSM solutions.

3.4. Governance arrangements and sustainability

Based on Table 1 and the challenges identified in Table 2, we found that the main influential factors of SSM implementation can be divided into three typical governance arrangements: Hierarchical, Closed co-governance and Open co-governance arrangement. The Self-governance arrangement does not seem to be relevant in implementation of SSM, since as mentioned in section 3.1.1, local authorities need to play a leading role, and in section 3.1.3, the difficulty in engaging stakeholders is described. For each arrangement, the most prominent actors, resources, rules of the game as well as discourses are listed and summarized in Table 3. Many differences exist between hierarchical and closed co-governance arrangements and an open co-governance arrangement, e.g., an increase of stakeholders' participation, shared-costs, knowledge and responsibilities between local authorities and stakeholders, and negotiations of decision making. With new legislation making a higher motivation to implement SSM must be expected. It is reasonable to argue that open co-governance arrangement is more useful for SSM implementation and private stakeholders' participation is the main influence factor (Bos and Brown, 2015; Brown et al., 2016; Chui and Ngai, 2016; Church, 2015; Cousins, 2017a,b; Dhakal and Chevalier, 2016; Dillon et al., 2016; Gao et al., 2016; Green et al., 2012; Herringshaw et al., 2010; Kaplowitz and Lupi, 2012; Lu et al., 2013; Newburn and Alberini, 2016; Nickel et al., 2014; Prosser et al., 2015; Thurston

Table 3
Characteristics of governance arrangements summarized from the literature.

	Hierarchical governance	Closed co-governance	Open co-governance
Actors	<i>Mainly government actors:</i> Water Department (water professionals)	<i>Selected mixed group of actors:</i> - Water Department - Planning Department (urban planners) - Parks Department (landscape architects) - Researchers - Contract Developers	<i>Large mixed groups of actors:</i> - Water, Planning, Parks Department - Researchers - Contract Developers - Private and NGO's stakeholders
Resources	<i>Power with government:</i> - Costs covered by the government - Public land - Lack of knowledge of SSM	<i>Power Pooled:</i> - Costs mainly covered by the government - Mainly public land - Shared knowledge of SSM among mixed group of actors	<i>Power Diffused:</i> - Costs shared by the government and private and NGOs stakeholders - Public land and private land - Shared and combined knowledge of SSM among a mixed group of stakeholders
Rules of the game	<i>Government coercion:</i> - The government makes decisions and enforces them - No legislative support for SSM	<i>Restricted cooperation:</i> - Joint decision making - No legislative support for SSM	<i>Flexible collaboration:</i> - Open decision making among a mixed groups of stakeholders - Legislative support to be provided
Discourses	<i>Risk of implementing SSM:</i> - Lack of awareness of social, economic, environmental benefits - Lack of trust - Unclear responsibility - Resistance to implementation of SSM	<i>SSM as an auxiliary solution:</i> - Some awareness of social, economic, environmental benefits - Openness among actors - Unclear responsibility - Some insecurity in relation to implementation of SSM	<i>SSM as an alternative solution:</i> - Awareness of social, economic, environmental benefits - Trust among actors - Clear shared responsibility - Motivation to implementation of SSM

et al., 2010). However, private stakeholders participation is a big challenge as mentioned in section 3.1.3. More focus on environmental sustainability rather than social and economic sustainability could be an explanation for lack of private stakeholder engagement (Cettner et al., 2014a,b; Chaffin et al., 2016; Church, 2015).

All the searched papers included the word “sustainable” or “sustainability”, but did not explain the meaning of “sustainable” or “sustainability” in detail. Based on the content, the papers can be categorized into two ‘sustainability’ types: Type 1 General Research of SSM (31 papers) and Type 2 Stakeholders Participation (13 papers). In Type 1, environmental sustainability is the primary aim of SSM implementation, such as reducing water pollution and pluvial flooding, although social and economic sustainability is mentioned as providing ancillary benefits. However, Type 2 papers with focus on private stakeholders have a broader focus related to social (aesthetic and recreation, etc.) and economic (shared costs or stormwater impact fee, etc.) (e.g., Bos and Brown, 2015; Gao et al., 2016; Lu et al., 2013). Also, in Type 2 papers, social and economic sustainability is primarily studied to increase private stakeholders' participation. It shows that the financial incentives can greatly increase the willingness to pay for SSM infrastructures (Chui and Ngai, 2016; Newburn and Alberini, 2016). In addition, projects involving stakeholders personal costs (land, money, time, etc.) can increase the private stakeholders participation, and also helps to build the trust among the actors (Bos and Brown, 2015; Brown et al., 2016; Dillon et al., 2016). Therefore, it is reasonable to argue that in the future, more research on increasing SSM implementation should focus on the social and economic sustainability.

4. Conclusions

Sustainable stormwater management (SSM) by using NbS-green infrastructure can reduce the risk of pluvial flooding and bring multi-benefits to human lives. In this paper, a literature review identified the challenges influencing the implementation of SSM from a governance perspective, which was based on four aspects: actors, resources, rules of the game, and discourses. Several critical factors emerged. First, there seems to be a general lack of evidence of SSM efficiency because of a general lack of knowledge related to

SSM. Therefore, standards and guidelines of SSM are lacking, which suggests that implementation of SSM is not incorporated into legislation. This may be the reason for unclear ownership, leadership, and distribution of responsibilities in large SSM projects. Second, lack of sufficient funding and sufficient space are frequently mentioned as obstacles to implementing SSM. Use of vacant land can partly solve the challenge of lack of space and the involvement of stakeholders can contribute with private investments and private land. Third, communication between stakeholders and local authorities is often mentioned as a problem because of mutual mistrust. Researchers are suggested to act as knowledge brokers to improve collaboration and cooperation among actors within the local authority, and between the local authority and local stakeholders, and also to play a major part in future large-scale implementation of SSM by supplying research-based evidence. On the local scale, we suggest a context-dependent open co-governance arrangement as a suitable way forward in implementing SSM. It is worth noting that more research on social-economic sustainability of SSM is needed, since it influences the private and non-governmental stakeholders' participation. The findings in this paper can help decision makers, water professionals, urban planners, landscape architects, researchers, and stakeholders by improving their understanding of the challenges to increase the implementing SSM and by improving general knowledge of SSM.

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